Preface to OCP Java SE 21 Developer (Exam 1Z0-830)

Part I: **OCP Java SE 21 Developer** of this book provides a comprehensive coverage of *new topics* that have come into Java 21 since the release of Java 17 and that are relevant for the *Java SE 21 Developer Professional Exam (1Z0-830)*. Primarily three new topics in Java 21 (*pattern matching for objects*, *sequenced collections*, and *virtual threads*) are included in the objectives for the new exam. Apart from that, the objectives for the *Java SE 21 Developer Professional Exam* did not change significantly from the objectives for the *Java SE 17 Developer Exam*.

Part II: **OCP Java SE 17 Developer** of this book provides extensive coverage for the following Java certifications:

- *Oracle Certified Professional: Java SE 17 Developer*, and its required *Java SE 17 Developer Exam (1Z0-829)*
- *Oracle Certified Professional: Java SE 11 Developer*, and its required *Java SE 11 Developer Exam (1Z0-819)*

Part I and **Part II** together provide a comprehensive resource for Java 21 certification:

- *Oracle Certified Professional: Java SE 21 Developer*, and its required *Java SE 21 Developer Professional Exam (1Z0-830)*

In this preface we provide the necessary details on how best to prepare and pass the *Java SE 21 Developer Professional Exam* using this programmer's guide. The syllabus for this exam is defined by a set of *exam objectives* that are presented in **Appendix A** in **Part I**. The *new* topics that have been added to the *Java SE 21 Developer Professional Exam* since Java 17 are covered in **Part I**, and **Appendix A** provides detailed references where the rest of the topics are covered in **Part II**: **OCP Java SE 17 Developer**.

About Part I: OCP Java SE 21 Developer

Our approach to writing **Part I** has not changed from the one we employed for our previous books, mainly because it has proved successful. The emphasis remains on analyzing code scenarios, rather than esoteric syntax of individual language constructs. The exam

continues to require actual experience with the language, not just mere recitation of facts. We still claim that proficiency in the language is the key to success.

Part I is no different from our previous books in one other important aspect: It provides a mixture of theory and practice that enables readers to prepare for the exam.

All elements found in our previous books (e.g., examples, figures, tables, review questions, mock exam questions) can be found in this one as well. We continue to use UML (*Unified Modeling Language*) extensively to illustrate concepts and language constructs, and all numbered examples continue to be complete Java programs ready for experimentation.

Each topic is explained and discussed thoroughly with examples, and is backed by review questions to reinforce the concepts. **Part I** is not biased toward any particular platform, but provides platform-specific details where necessary.

Part I is primarily intended for professionals who want to prepare for the *Java SE 21 Developer Professional Exam*, but it is readily accessible to any programmer who wants to master the new topics. After mastering the language and the core APIs by working systematically through **Part I** and **Part II**, the reader can confidently sit for the exam.

Part I has a separate appendix (**Appendix A**) providing all the pertinent information on preparing for the exam. Since **Part I** only covers new topics that are relevant for the Java 21 exam, it should be used with **Part II**, which covers the remaining exam topics.

The table of contents; listings of tables, examples, and figures; and a comprehensive index facilitate locating topics discussed in **Part I**. Cross-references are provided where necessary to indicate the relationships between the various features of Java.

In particular, we draw attention to the following features of **Part I**:



Chapter Topics

Each chapter starts with a short summary of the topics covered in the chapter, pointing out the major concepts that are introduced.



Prerequisites

Each chapter starts with a short summary of topics that are prerequisites for the topics covered in the chapter. **Part II**: **OCP Java SE 17 Developer** readily provides coverage for these prerequisites.

Exam Objectives

Developer Exam Objectives	
[0.1]	Exam objectives that are covered in each chapter are stated clearly at the beginning of every chapter.
[0.2]	The number in front of the objective identifies the exam objective, as defined by Oracle. The objectives are organized into major sections, detailing the curriculum for the exam.
[0.3]	The objectives for the *Java SE 21 Developer Professional Exam* are reproduced verbatim in **Appendix A**. This appendix also maps each exam objective to relevant chapters and sections in **Part I** and in **Part II**: **OCP Java SE 17 Developer**.

Supplementary Topics
• Supplementary topics are Java topics that are *not* on the exam per se, but which the candidate is expected to know. • Any supplementary topic is listed as a bullet at the beginning of the chapter.



Review Questions

Review questions are provided after every major topic to test and reinforce the material. The review questions predominantly reflect the kind of multiple-choice questions that can be asked on the actual exam. On the exam, the exact number of answers to choose for each question is explicitly stated. The review questions in **Part I** follow that practice.

Many questions on the actual exam contain code snippets with line numbers to indicate that complete implementation is not provided, and that the necessary missing code to compile and run the code snippets can be assumed. The review questions in **Part I** pro-

vide complete code implementations where possible, so that the code can be readily compiled and run.

Annotated answers to the review questions are provided in **Appendix B** in **Part I**.

Example 0.1 *Example Source Code*

We encourage readers to experiment with the code examples to reinforce the material in the book. The source code for the examples can be downloaded from the website for **Part I** (see **p. xviii**), and readily imported into the Eclipse IDE.

Java code in the book is presented in `a monospaced font`. Lines of code in the examples or in code snippets are referenced in the text by a number, which is specified by using a single-line comment in the code. For example, in the following code snippet, the call to the method `doSomethingInteresting()` at (1) does something interesting:

Click here to view code image

```
// ...
doSomethingInteresting();                    // (1)
// ...
```

Names of classes and interfaces start with an uppercase letter. Names of packages, variables, and methods start with a lowercase letter. Constants are in all uppercase letters. Interface names begin with the prefix `I`, when it makes sense to distinguish them from class names. Coding conventions are followed, except when we have had to deviate from these conventions in the interest of space or clarity.

Java SE Platform API Documentation

fully.qualified.Name

A vertical gray bar is used to highlight methods and fields found in the classes of the Java SE Platform API.

Any explanation following the API information is also similarly highlighted.

To get the maximum benefit from using **Part I** in preparing for the *Java SE 21 Developer Professional Exam*, we strongly recommend installing the latest version (Release 21 or newer) of the JDK and its accompanying API documentation. **Part I** focuses solely on the new topics that were finalized in Java SE 21 since the release of Java SE 17.

Website for Part I: OCP Java SE 21 Developer

Part I is backed by a website:

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https://www.mughal.no/jse21ocp/
```

Auxiliary material on the website includes the following:

- Source code for all the examples in **Part I**
- Annotated answers to the reviews questions in **Part I**
- Table of contents, sample chapters, and index from **Part I**
- Content specific for the *Java SE 21 Developer Professional Exam*
- Errata for **Part I**
- Links to miscellaneous Java resources (e.g., certification, discussion groups, and tools)

Information about the Java Standard Edition (SE) and its documentation can be found at the following website:

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www.oracle.com/technetwork/java/javase/overview/index.html
```

The current authoritative technical reference for the Java programming language, *The Java® Language Specification: Java SE 21 Edition*, can be found at this website:

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http://docs.oracle.com/javase/specs/index.html
```

Request for Feedback

Considerable effort has been made to ensure the accuracy of the content of this book. All code examples (including code fragments) have been compiled and tested on various platforms. In the final analysis, any errors remaining are the sole responsibility of the principal author.

Any questions, comments, suggestions, and corrections are welcome. Let us know whether the book was helpful (or not) for your purpose. Any feedback is valuable. The principal author and the co-author can be reached at the following e-mail addresses, respectively:

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Khalid A. Mughal is the principal author of this book, primarily responsible for writing the material covering the Java topics. He is also the principal author of several other books on previous versions of the Java certification exam: Java SE 17 OCP (1Z0-829), Java SE 11 OCP (1Z0-819), Java SE 8 OCA (1Z0-808), Java SE 6 (1Z0-851), and SCPJ2 1.4 (CX-310-035).

Khalid is an associate professor emeritus at the Department of Informatics at the University of Bergen, Norway, where he was responsible for designing and implementing various courses in informatics. Over the years, he has taught programming (primarily Java), software engineering (object-oriented system development), databases (data modeling and database management systems), compiler techniques, web application development, and software security courses. For 15 years, he was responsible for developing and running web-based programming courses in Java, which were offered to off-campus students. He has also given numerous courses and seminars at various levels in object-oriented programming and system development using Java and Java-related technologies, both at the University of Bergen and various other universities in Norway and East Africa, and also for the IT industry.

Vasily A. Strelnikov

Vasily Strelnikov is primarily responsible for developing new review questions for the chapters contained in this book.

Vasily is a senior principal OCI (Oracle Cloud Infrastructure) solution specialist, working at Oracle for more than 27 years. He is a co-author of the Java EE 7, Java SE 8, Java SE 11, Java SE 17, and Java SE 21 Certification exams. He has designed multiple Java courses that are offered by Oracle: Java SE 21 Developer, Java SE 17 Programming Complete, Java SE 11 Programming Complete, and Java SE 8 Certification Preparation Seminar. He has also created the Developing Applications for the Java EE 7 Platform training at Oracle. Vasily has over 25 years of experience in Java. He specializes in Java middleware application development and web services.

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For the technical review of **Part I**, we were again lucky to have a Java guru who graciously agreed to take on the task:

Mikalai Zaikin is a lead Java developer at IBA Lithuania, and is currently located in Vilnius. He has helped Oracle with development of Java certification exams and has also been a technical reviewer of several Java certification books. He also contributes to the Java Quiz column for Oracle's *Java Magazine* in collaboration with Simon Roberts.

Without doubt, Mikalai has an eye for detail. It is no exaggeration to say that his feedback has been invaluable in improving the quality of **Part I** at all levels. Our most sincere thanks to Mikalai for the many excellent comments and suggestions on the contents—especially regarding code examples and review questions, and above all, for weeding out numerous pesky errors in the manuscript.

Great effort has been made to eliminate mistakes and errors in this book. Any remaining oversights are solely our responsibility. We hope that our readers will bring them to our attention when they find them.

Family support was undoubtedly essential in this project as well and for that we are very grateful to our families for putting up with us.

We wish our reader all the best in going down the caffeine-infused path of Java certification. May your threads stay untangled and complete normally!

—Khalid A. Mughal
November 5, 2024
Bergen, Norway

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